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Studierichting: Informatica
Oefeningenreeks 5D nr. 3.3

$$\begin{aligned}f(x) - g(x) &= x^2 + 4x - x + 2 \\&= x^2 + 3x + 2\end{aligned}$$

Snijpunten zoeken: $x^2 + 3x + 2 = 0$

Som en product van wortels:

$$\text{Som: } x_1 + x_2 = -\frac{b}{a} = -\frac{3}{1} = -3$$

$$\text{Product: } x_1 \cdot x_2 = \frac{c}{a} = \frac{2}{1} = 2$$

Zoeken naar twee getallen met som -3 en product 2 :

$$\Rightarrow x_1 = -1, x_2 = -2$$

Oppervlakte bepalen per interval:

$$\int_{-3}^{-2} (x^2 + 3x + 2)dx + \int_{-2}^{-1} (x^2 + 3x + 2)dx + \int_{-1}^0 (x^2 + 3x + 2)dx$$

$$\text{Primitieve: } \int (x^2 + 3x + 2)dx = \frac{x^3}{3} + \frac{3x^2}{2} + 2x$$

$$= \left[\frac{x^3}{3} + \frac{3x^2}{2} + 2x \right]_{-3}^{-2} - \left[\frac{x^3}{3} + \frac{3x^2}{2} + 2x \right]_{-2}^{-1} + \left[\frac{x^3}{3} + \frac{3x^2}{2} + 2x \right]_{-1}^0$$

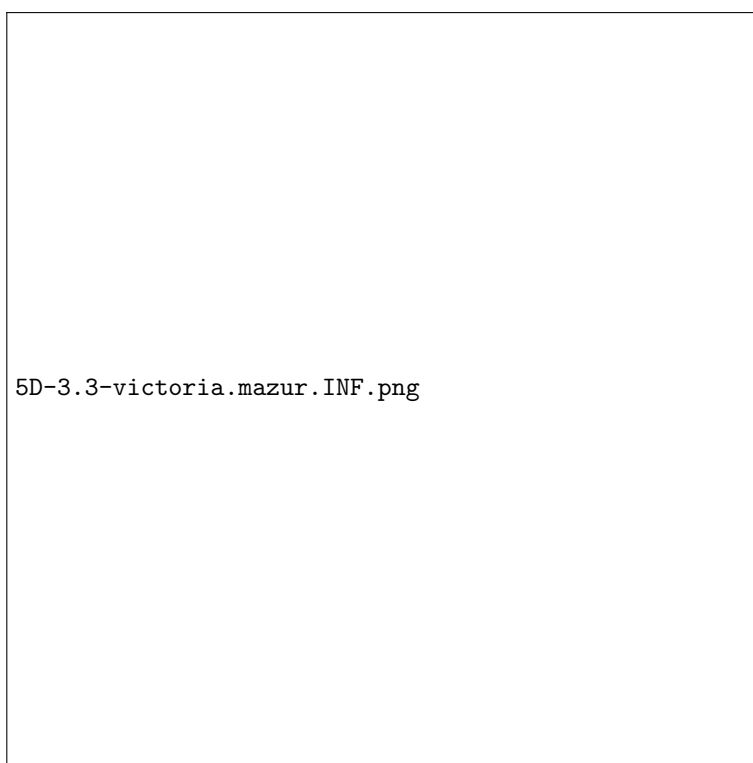
Evaluëren van grenzen:

$$\text{Bij } x = -2: \frac{(-2)^3}{3} + \frac{3(-2)^2}{2} + 2(-2) = -\frac{8}{3} + 6 - 4 = -\frac{8}{3} + 2 = \frac{-8 + 6}{3} = -\frac{2}{3}$$

$$\begin{aligned}\text{Bij } x = -3: \frac{(-3)^3}{3} + \frac{3(-3)^2}{2} + 2(-3) &= -9 + \frac{27}{2} - 6 = -15 + \frac{27}{2} = \\ \frac{-30 + 27}{2} &= -\frac{3}{2}\end{aligned}$$

$$\begin{aligned}&= -\left(-\frac{2}{3} + \frac{3}{2}\right) - \left(-\frac{5}{6} + \frac{3}{3}\right) - \left(-\frac{5}{6}\right) \\&= \frac{11}{6}\end{aligned}$$

References



5D-3.3-victoria.mazur.INF.png